PATIENT CARE THEORY 2

UNIT 7, PART 3: Pediatric Trauma & CVS

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Learning Objectives

- Discuss trauma and shock in the pediatric patient
- Identify cardiovascular emergencies seen in pediatric patients
  - Bradydysrhythmias/tachydysrhythmias

Epidemiology of Trauma

- Trauma is the leading cause of death in patients >1 yr.
- 25% of traumatic injuries occur in children
  - Traumatic brain injury
  - MVC
  - Submersion injuries
  - Homicide
  - Suicide
  - Fires

Epidemiology

- Teens are at increased risk due to exposure to hazards
- Infants are at greater risk of inflicted trauma

Pediatric Trauma Score

Assessment Component	SCORE		
	+2	+1	-1
Weight	Weight >20 kg (>44 lb)	10-20 kg (22-44 lb)	<10kg (<22 lb)
Airway	Normal	Oral or nasal airway, oxygen	Intubated, cricothyroidotomy, or tracheostomy
Systolic Blood Pressure	>90 mm Hg, good peripheral pulses and perfusion	50-90 mm Hg, carotid/femoral pulses palpable	<50 mm Hg, weak or no pulses
Level of Consciousness	Awake	Obtunded or any loss of consciousness	Coma, unresponsive
Fracture	None seen or suspected	Single, closed	Open or multiple
Cutaneous	None visible	Contusion, abrasion, laceration <7 cm not through fascia	Tissue loss, any gunshot wound or stab wound through fascia
Totals			

Adapted with permission from Tepas JJ, Molitt DL, Talbert JL, et al: The pediatric trauma score as a predictor of injury severity in the injured child. Journal of Pediatric Surgery. 1987;22(1)15.

*PTS > 8 should have 0 % mortality.

All injured children with PTS < 8 should be triaged to an appropriate pediatric trauma center.

Circulation

- Hypovolemia is the most common cause of shock in children worldwide (P.A.L.S.)
- Smaller blood vol. → children loose large % blood quickly, but also compensate with minimal changes to BP (BP may initially be normal with significant hypovolemia)
- Followed by sudden deterioration
- Fluid (crystalloid) up to 3x 20ml/kg bolus
- Always consider non-obvious signs of bleeding

Hypovolemia

- Rapid control of external bleeding essential due to small blood volume
- Efficient compensation makes recognition of shock difficult
- Kids get sick differently than adults
- Sudden decompensation, onset of irreversible shock may occur
  - i.e. unlike an adult, a child may maintain a BP and HR at a steady state and then very suddenly deteriorate
  - Unlike adults where we typically see a steady decline in BP and increase in HR

Hypovolemia

- BP monitoring = Poor shock indicator
- Assess perfusion using:
  - Peripheral pulse rate, quality
    - tachycardia
  - *Skin colour*, temperature
  - LOC
  - Capillary refill > 2 seconds

Hypovolemia

Management

- ABCDE
- Minimal scene time
- Rapid assessment
- Stop any bleeding and maintain warmth
- CPR?

Enroute

- For hypotension…
  - IV access & fluid resuscitation 20 cc/kg of NS
  - Keep in mind though they need blood…..

Shock

- Hypovolemia is the most common cause of shock in children worldwide (P.A.L.S.)

Frequently results from:

- Hypovolemia from sudden hemorrhagic volume loss
- Burns (slower loss of fluids)
- Diarrhea
- Vomiting
- Diabetic ketoacidosis**
- Poisoning

Fluid Therapy recommendations in Pediatrics

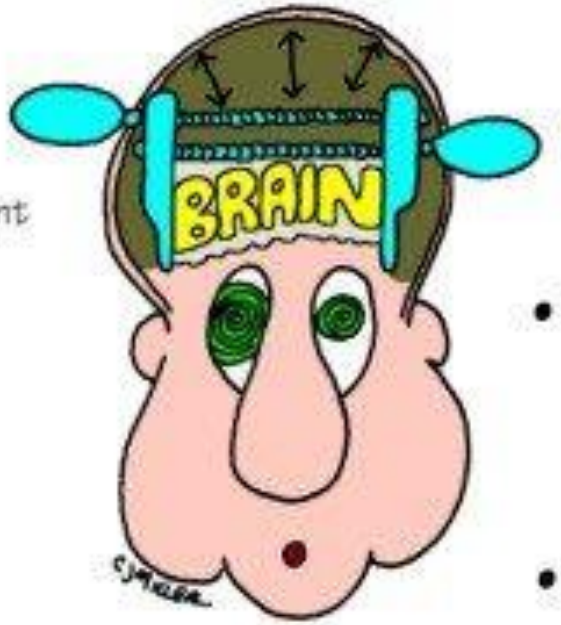
- In general: ***Hypovolemic*** Shock ;20 ml/kg over 5-10 min
  - If not blood loss – more fluid may be required
- ***Cardiogenic*** shock – 5-10ml/kg given over 10-20 minutes (monitoring for ACPE)
- ***Septic*** Shock – 20ml/kg to start; may require up to 60ml/kg (patch the BHP for orders)
- Diabetic Ketoacidosis – Patch required 10-20 ml/kg given over 60-120 minutes

TBI

- Head trauma is the leading cause of death and disability in childhood (20-30% mortality rate)
- Causes
  - Infancy: NAT (non-accidental trauma), falls
  - Childhood: MVA, pedestrian struck, bicycle
  - Adolescence: MVA, pedestrian, bikes, boards, violence
- Incidence is equal between male and female 0-5yr: after 5 yrs males are more likely to sustain a head injury

INCREASED INTRACRANIAL PRESSURE

- Changes in LOC
- Eyes
  - Papilledema
  - Pupillary Changes
  - Impaired Eye Movement
- Posturing
  - Decerebrate
  - Decorticate
  - Flaccid
- Decreased Motor Function
  - Change in Motor Ability
  - Posturing



- Headache
- Seizures
  - Impaired Sensory & Motor Function
- Changes in Vital Signs:
  - Cushing's Triad:
    - ↑ Systolic B/P
    - ↓ Pulse
    - Altered Resp Pattern
- Vomiting
- Changes in Speech
- Infants:
  - Bulging Fontanels
  - Cranial Suture Separation
  - ↑ Head Circumference
  - High Pitched Cry

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TBI

Management

- Elevate level of the head (30 °)
- Prevent hypoxia and hypotension
  - Hypotension based on age?
  - BVM rates?
- Control any external hemorrhage
- Maintain SMR (c-spine)

Chest Trauma

- 4.5 – 8% of pediatric trauma
- 2nd most common cause of mortality
- Causes:
  - MVA
  - Pedestrians
  - Unrestrained passengers
  - Bicycle riders
  - Falls

Chest/Thoracic Trauma

Pediatric thoracic trauma higher risk due to:

- More compliant chest wall
- Increased mediastinal mobility
  - Tension pneumo develops more quickly
- More prone to hypoxia
- Decreased ability to compensate for hypovolemia

Chest/Thoracic Trauma

Blunt vs Penetrating

- Blunt:
  - High energy trauma affects the internal organs
  - Deceleration mechanisms affect mediastinal structures
- Penetrating:
  - Less protection from chest wall
  - Do not remove penetrating objects!

Abdominal Trauma

- What organs are we concerned about?

Abdominal Trauma

- 3rd leading cause of traumatic death
- Often missed in children

Sign	Possible Injuries
Seatbelt Injury	Small bowel injury Chance fracture
Handlebar injury	Duodenal hematoma Pancreatic injury
Sport related injury	Spleen, kidney, bowel

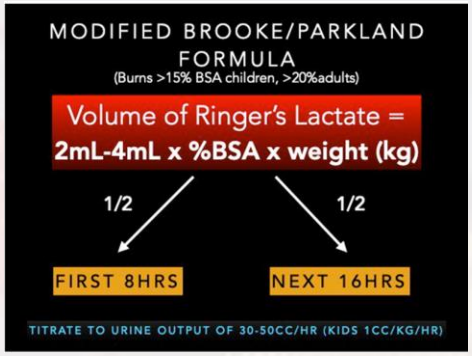
Burns

- 70% of ped burns secondary to hot liquids
- Up to 20% of burns in younger children are secondary to abuse and neglect
- Thermal injury = loss of fluid through the burns
  - and into adjacent tissues (third space loss)
- Fluid loss is greatest in the first 12-36 hours

Burns

Management

- Use universal precautions
  - Wear mask & gloves to protect the patient
- Manage burn areas as per BLS Standards
- ABCDE
- ANALGESIA
- Volume depletion takes hours
  - Fluid resuscitation?
  - IV & 20cc/kg if blood loss suspected
  - For burn related fluid loss – modified Brooke/Parkland



Submersion Injuries

- “Drowning is the process of experiencing respiratory impairment from submersion/immersion in liquid. Drowning outcomes are classified as death, morbidity and no morbidity” (WHO)
  - adopted by the 2002 World Congress on Drowning
- Drowning = death within 24h of suffocation from submersion in liquid
- Near drowning/submersion injury = survival > 24hr past the event

Submersion Injuries

- >50% of drowning victims are <5 yr/o (peak at 1-2 yr/o)
  - Fatalities are highest in this group
  - Males > Females
- Most common cause of cardiac arrest in children
  - <1 yr: bathtubs (55%)
  - 1-4 yrs: pools (56%)
  - >4 yrs freshwater (63%)

Submersion Injury

Patho

- Accidental submersion
- Loss of normal breathing pattern
- Possible laryngospasm
- Aspiration/breath-holding/apnea
- Hypoxemia
  - Low O2, hypercarbia, acidosis
  - End organ damage

Submersion injury

Management

- ABCDE
- Consider trauma/c-spine (0.5% have C-spine injuries)
- Remove wet clothing
- Expect/treat hypothermia

Poor Prognosis:

  - Prolonged submersion >25 min
  - Delay in CPR
  - Resuscitation > 25 min
  - Dilated and Fixed Pupils
- When is a drowning considered code 5?

- **Child Revived Almost 2 Hours After "Drowning" - But How?**
  - https://www.iflscience.com/health-and-medicine/child-survived-almost-2-hours-cpr-after-falling-icy-stream-how/
- How an Italian Boy Survived 42 Minutes Underwater
  - https://time.com/3897897/how-an-italian-boy-survived-42-minutes-underwater/

RHYTHM DISTURBANCES

Rhythms

In pediatric patients, rhythms are categorized as:

- Slow pulse - bradyarrhythmias
- Fast pulse – tachyarrhythmias
- No pulse – pulseless rhythms
  - The urgency of treatment for a slow or fast pulses depends on the rate, systemic perfusion and the patient’s overall clinical presentation

Bradyarrhythmias

- Hypoxemia is the most frequent cause of bradyarrhythmias in the pediatric patient
- A heart rate less than 100 in a newborn and less than 80 in a small child is very slow
- Look for signs of hypoperfusion
  - Altered LOC, agitation
  - Poor skin colour
  - Delayed capillary refill
  - Normal BP ($\geq$ 2 x age + 90)
  - Low BP (< 2 x age + 70)
  - etc

Bradyarrhythmias

Management

- ABCD
- Call for ACP backup
- Minimal scene time
- Rapid transport
  - If a slow rhythm persists despite adequate O_2 & PPV
  - Initiate transport

Tachyarrhythmias

- Sinus tachycardia (ST)
- Heart rate typically < 180

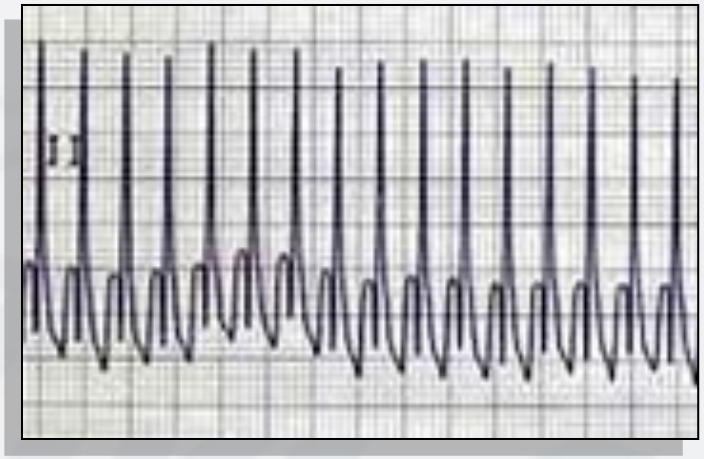
Causes

- Hypoxemia
- Hypovolemia
- Hyperthermia
- Metabolic stress
- Toxins/poisons/drugs
- Pain or anxiety

Management

- Treat the underlying cause

Tachyarrhythmias



SVT (QRS ≤0.08 sec.)

- AVNR is the most common cause
- Often occurs is paroxysms (PSVT)
- In infants, SVT generally produces heart rates > 220/min. and sometimes as high as 300/min.
- Most SVTs (60%) are > 230/min. (PALS)
- HR > 180 in children is SVT until proven otherwise

Tachyarrhythmias

Clinical presentation of SVT

- SVT may be initially well tolerated in most infants and older children
- Can also lead to acute CHF / collapse
- Clinical presentation depends on pt age, duration of the SVT
- Typical complaints
- Lightheadedness, dizziness
- Chest discomfort
- Delayed capillary refill

Tachyarrhythmias

SVT Management

- ABCD
- Call for ACP backup
- Minimal scene time
- Rapid transport

ST vs. SVT

ST

- Hx compatible with ST (e.g. fever, injury dehydration, pain)
- P wave present/normal
- Heart rate varies with activity
- Variable R-R with constant P-R
- Infants: HR usually < 220
- Children: HR usually < 180

SVT

- Hx not compatible with ST or non-specific
- P waves absent/abnormal
- HR does not vary with activity
- Abrupt rate changes
- Infants: HR usually > 220
- Children: HR usually > 180

Ventricular Tachycardia

- Uncommon in the paediatric age
- Most children with VT have an underlying structural heart disease, prolonged QT or myocarditis/cardiomyopathy
- Other causes include: acute hypoxemia, acidosis, electrolyte imbalance, toxins/drugs/poisons (e.g. tricyclic anti-depressants)
- ABCD
- Call for ACP backup
- Minimal scene time
- Rapid transport
- Pulseless VT is treated like VF with defibrillation (no patch)

MANAGEMENT SUMMARY

Management

- Supportive care
  - Ensure a patent airway
  - Hint of an A/W problem is life-threatening until proven otherwise
  - O_2 at high concentration
  - PPV prn
    - Hypoxia = common cause of morbidity and mortality
  - Control external bleeding
  - Suspect volume loss + hypotension – IV & fluids
  - Shock can be insidious – maintain a high index of suspicion (perfusion over hypotension)

Management

- Bradyarrhythmias are first managed by ensuring the airway is patent and the patient adequately oxygenated and ventilated
  - CPR (in children until showing signs of puberty)
    - HR < 60
    - Signs of hypoperfusion
  - Other options: ACP backup
- Tachyarrhytthmias
  - ACP backup

QUESTIONS